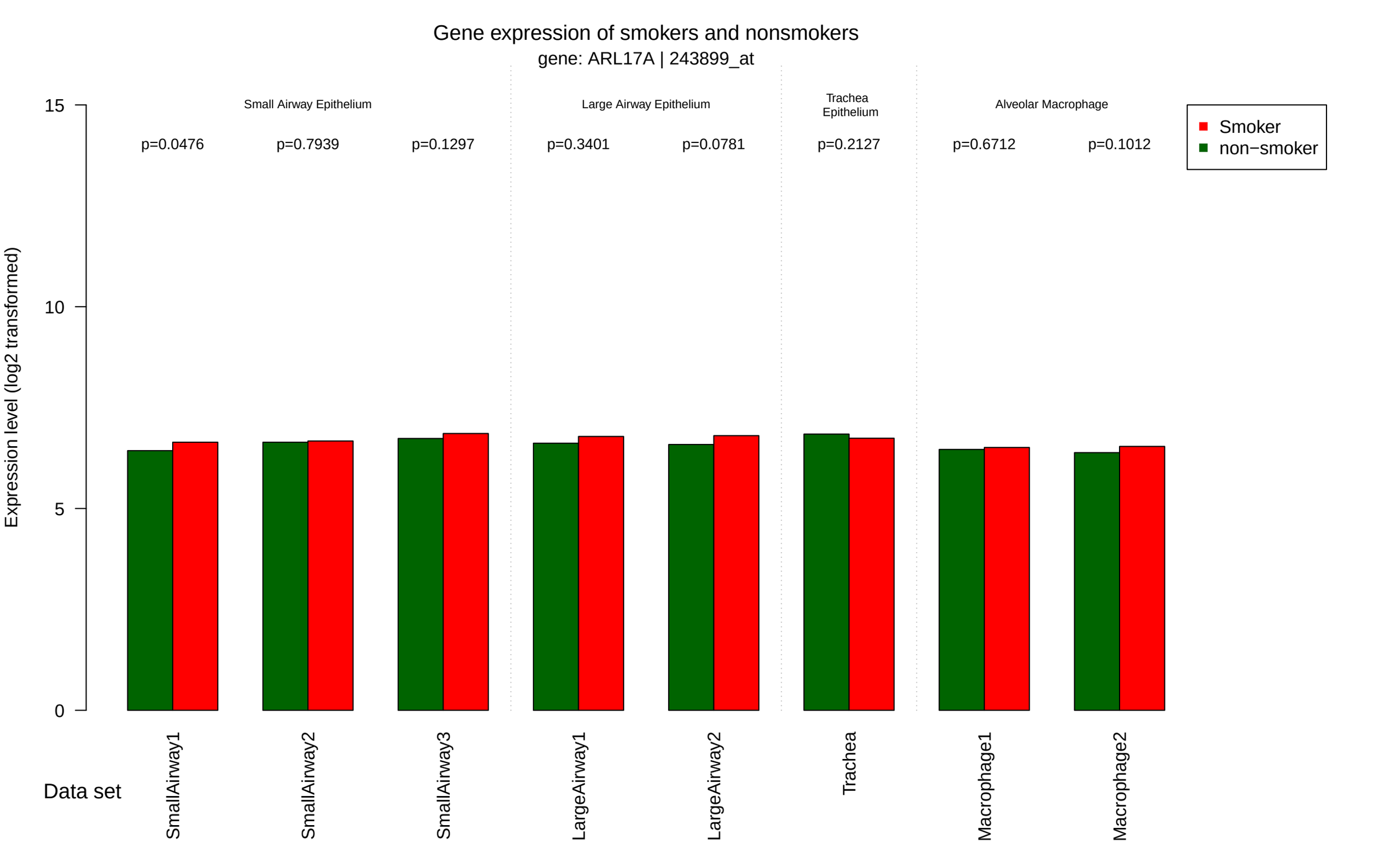
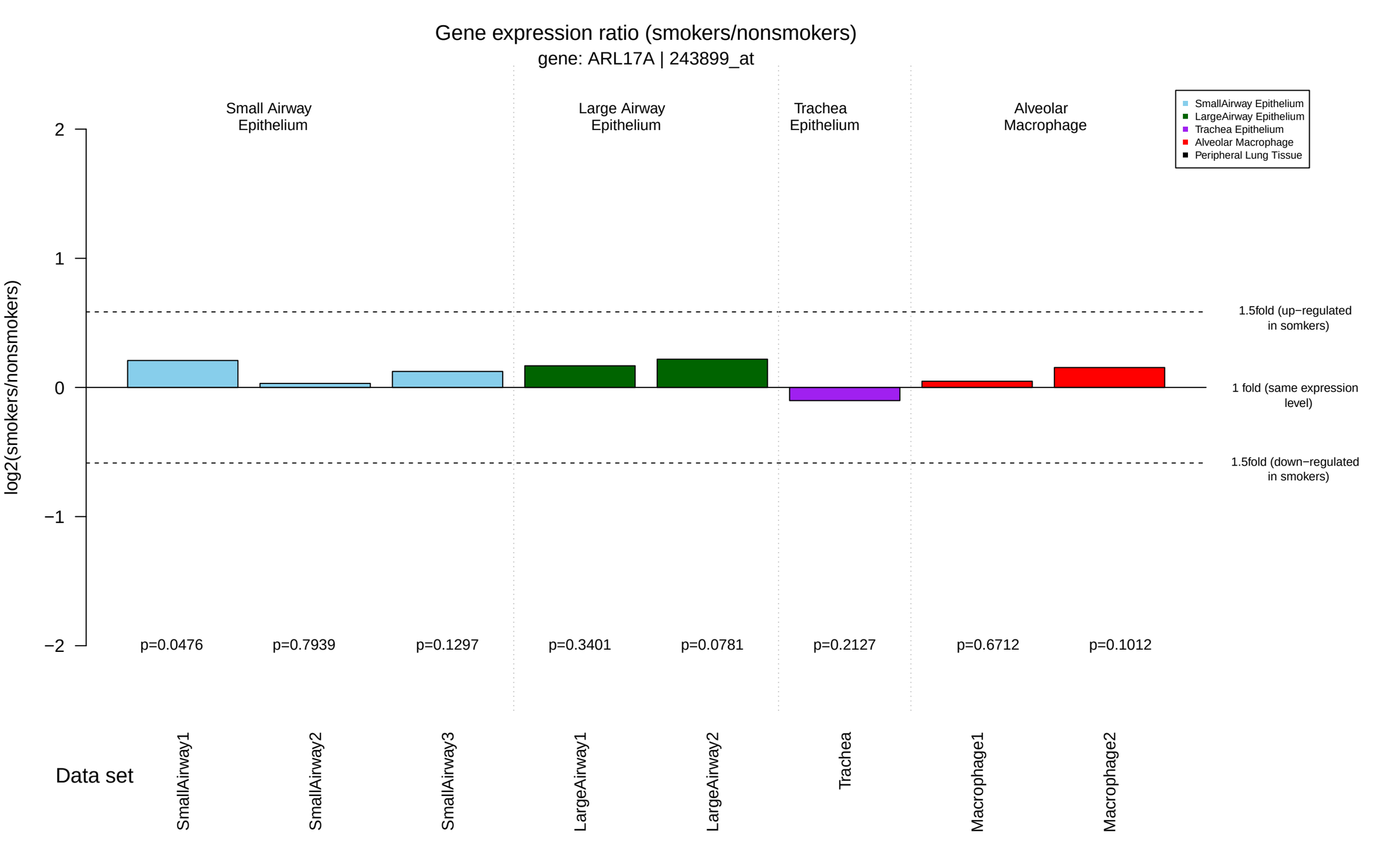
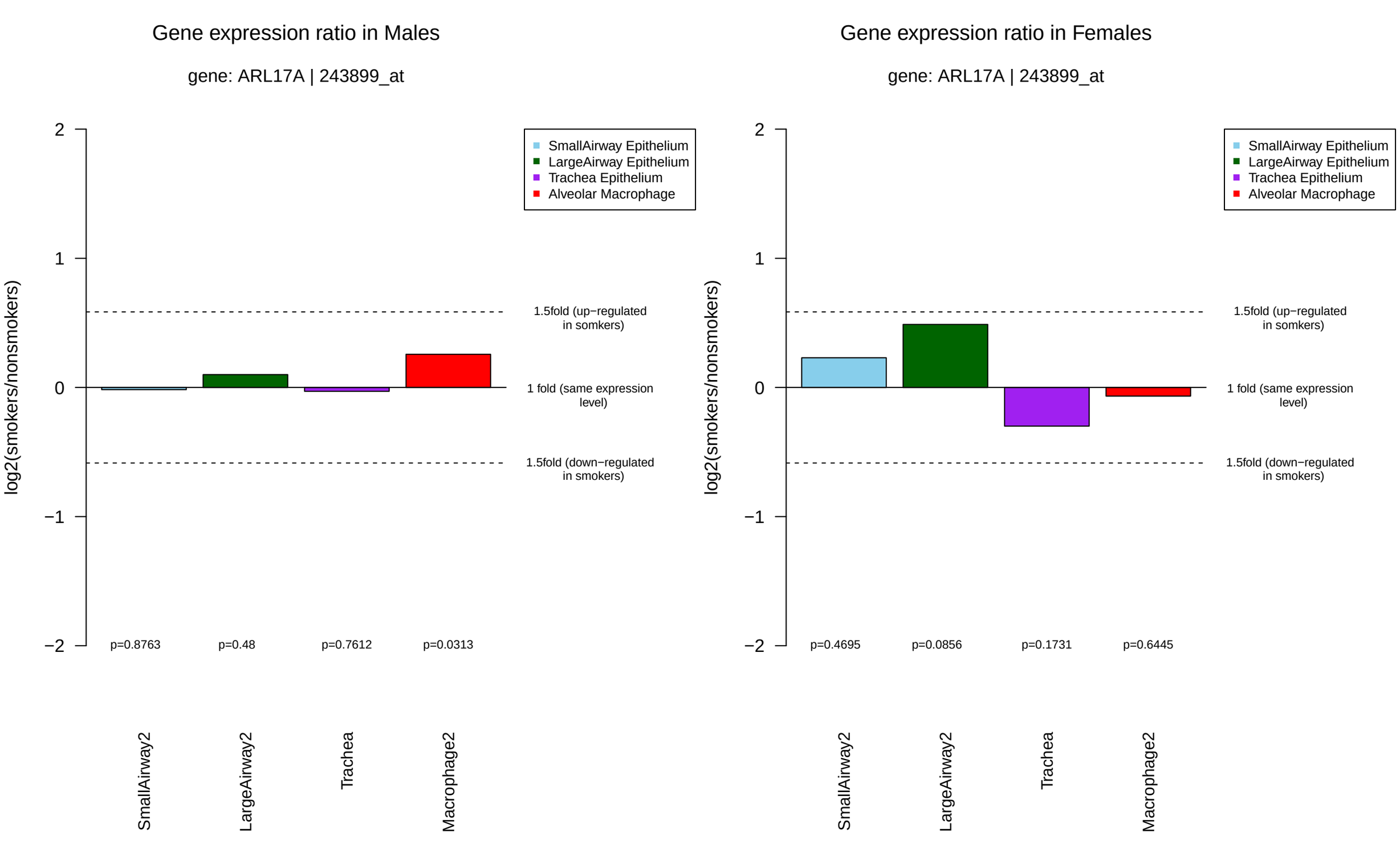
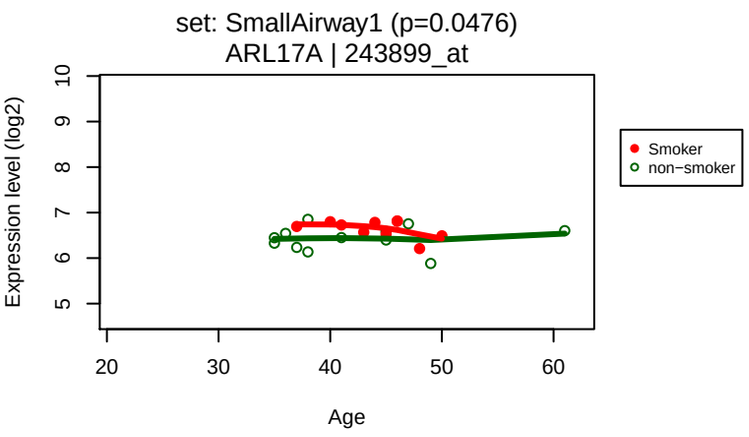
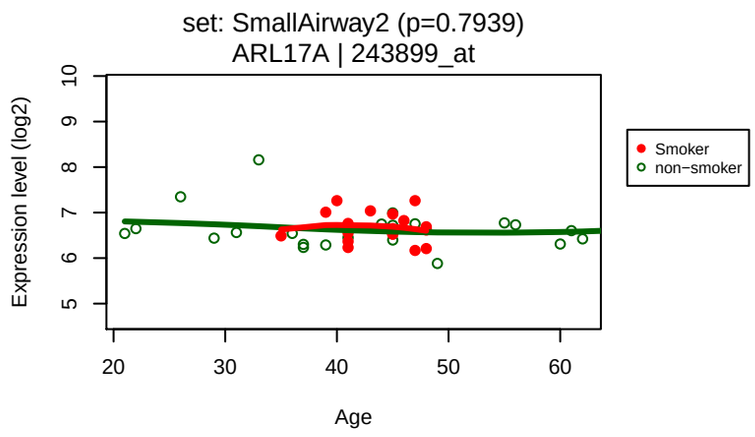
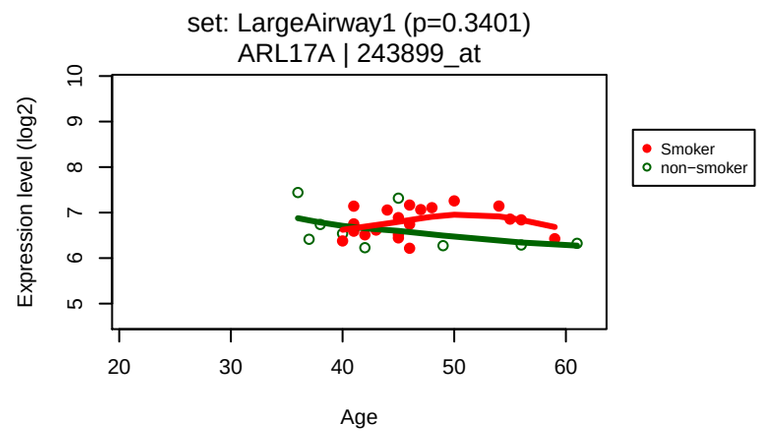

Gene expression of smokers and nonsmokers
gene: ARL17A | 243899_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0476
p=0.7939
p=0.1297
p=0.3401
p=0.0781
p=0.2127
p=0.6712
p=0.1012
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

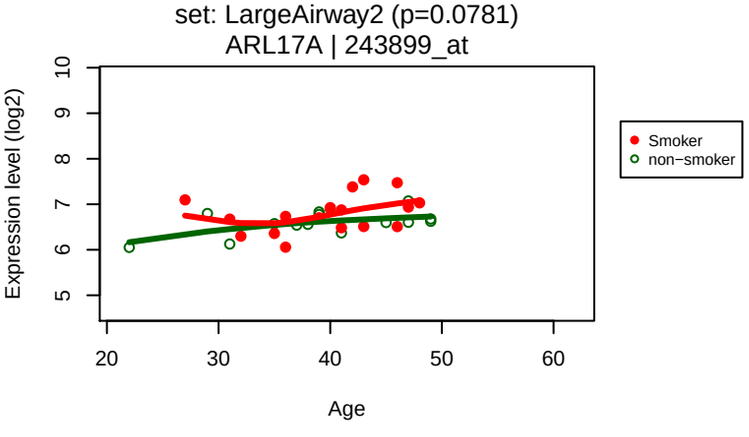

Gene expression ratio (smokers/nonsmokers)
gene: ARL17A | 243899_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0476
p=0.7939
p=0.1297
p=0.3401
p=0.0781
p=0.2127
p=0.6712
p=0.1012
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

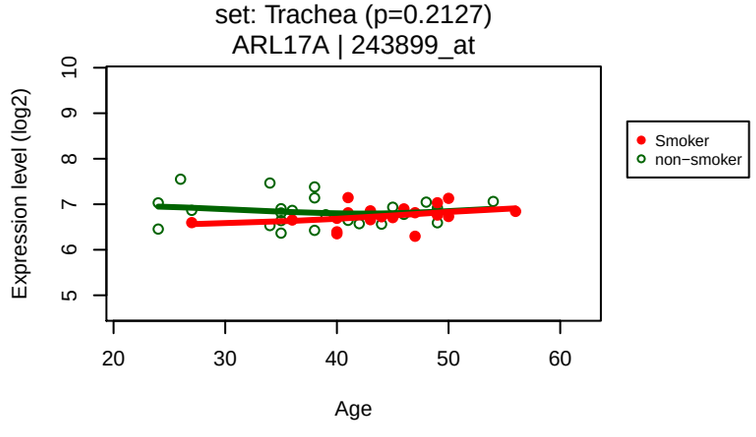

Gene expression ratio in Males
gene: ARL17A | 243899_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8763
p=0.48
p=0.7612
p=0.0313
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL17A | 243899_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4695
p=0.0856
p=0.1731
p=0.6445
SmallAirway2
LargeAirway2
Trachea
Macrophage2

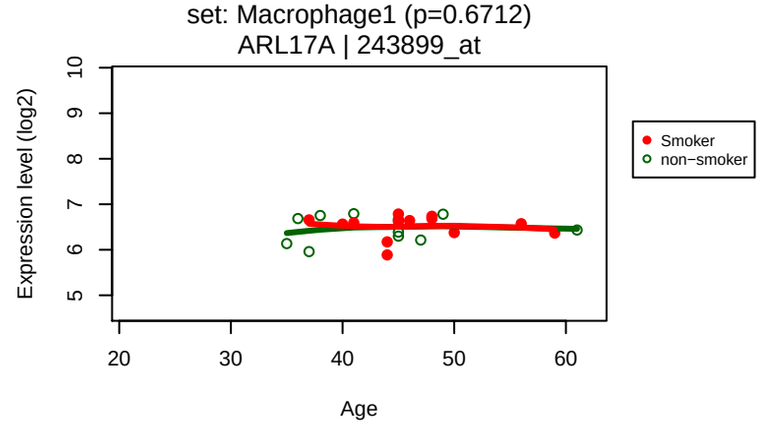

set: SmallAirway1 (p=0.0476)
ARL17A | 243899_at
Expression level (log2)
Age
Smoker
non−smoker

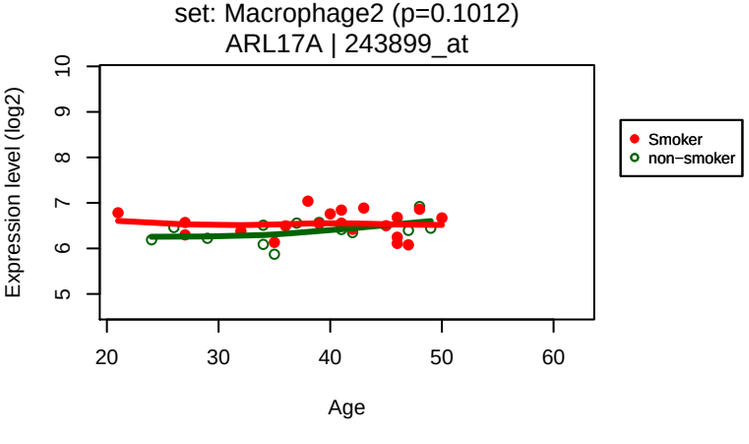

set: SmallAirway2 (p=0.7939)
ARL17A | 243899_at
Expression level (log2)
Age
Smoker
non−smoker

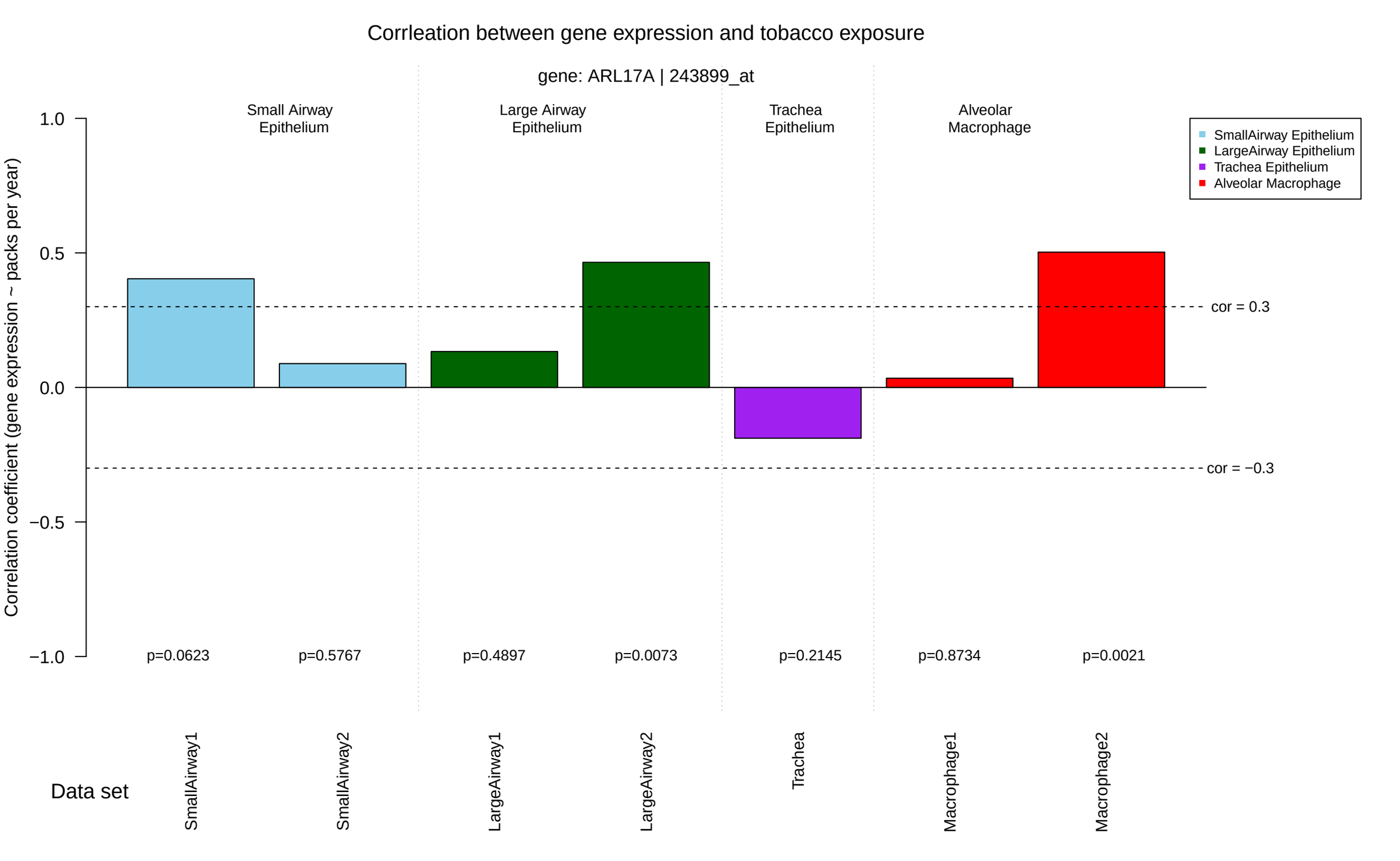

set: LargeAirway1 (p=0.3401)
ARL17A | 243899_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0781)
ARL17A | 243899_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2127)
ARL17A | 243899_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6712)
ARL17A | 243899_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1012)
ARL17A | 243899_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL17A | 243899_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0623
p=0.5767
p=0.4897
p=0.0073
p=0.2145
p=0.8734
p=0.0021
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2